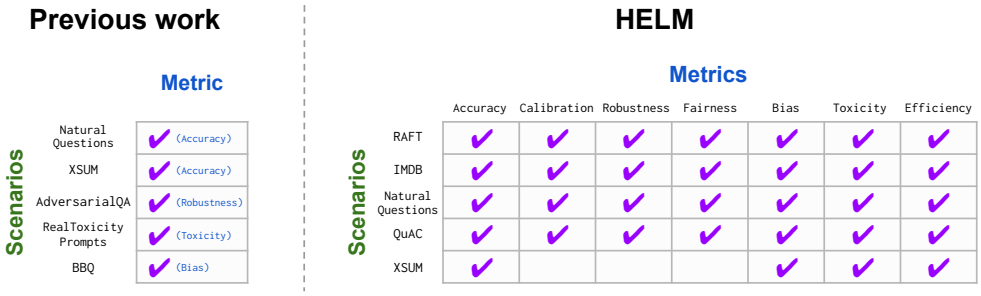

Figure 3: **Many metrics for each use case.** In comparison to most prior benchmarks of language technologies, which primarily center accuracy and often relegate other desiderata to their own bespoke datasets (if at all), in HELM we take a multi-metric approach. This foregrounds metrics beyond accuracy and allows one to study the tradeoffs between the metrics.

This multi-metric perspective conveys a position we take on evaluation practices in AI. While most benchmarks primarily foreground accuracy, perhaps deferring the evaluation of other metrics (e.g. the extent to which models generate toxic content) to separate scenarios (e.g. **RealToxicityPrompts**), we believe it is integral that all of these metrics be evaluated in the same contexts where we expect to deploy models (see Figure 3). In particular, measuring these 7 desiderata for the same scenarios makes explicit potential tradeoffs and helps to ensure these desiderata are not treated as second-class citizens to accuracy (see Friedman & Nissenbaum, 1996).

Targeted evaluations. In addition to our core set of 16 scenarios, where for each scenario we measure all 7 categories of metrics, HELM has 7 targeted evaluations through 26 additional scenarios and accompanying metrics. These evaluations target linguistic understanding, world and commonsense knowledge, reasoning capabilities, memorization and copyright, disinformation generation, biases, and toxicity generation, providing a deeper dive beyond the core scenarios. This includes 21 scenarios that are either entirely new (e.g. **WikiFact**) or that have not been used in mainstream language model evaluation (e.g. **ICE**). While HELM is oriented by a holistic approach that foregrounds societal impact and is reflected in our multi-metric perspective, evaluation can also pinpoint specific phenomena to advance scientific understanding (e.g. a model's ability to perform analogical reasoning; see Bommasani et al., 2021, §4.4). For this reason, to make our evaluation results more intelligible, we separate the core scenarios from the targeted evaluations: the core scenarios and multi-metric measurement provide an integrated lens on models, whereas the targeted evaluations isolate specific skills and risks.

Standardization. To build a shared understanding of existing language models, consistent with our third element of holistic evaluation, we benchmark 30 prominent language models on HELM. These models come from 12 organizations: AI21 Labs (e.g. J1-Jumbo v1 (178B)), Anthropic (Anthropic-LM v4-s3 (52B)), BigScience (e.g. BLOOM (176B)), Cohere (e.g. Cohere xlarge v20220609 (52.4B)), EleutherAI (e.g. GPT-NeoX (20B)), Google (e.g. UL2 (20B)), Meta (e.g. OPT (175B)), Microsoft/NVIDIA (e.g. TNLG v2 (530B)), OpenAI (e.g. davinci (175B)), Tsinghua University (GLM (130B)), and Yandex (YaLM (100B)). Benchmarking these models is challenging given they vary in accessibility (see Liang et al., 2022): some are open (e.g. GPT-NeoX (20B)), some are limited-access (e.g. davinci (175B)), and some are closed (e.g. Anthropic-LM v4-s3 (52B)). In some cases, very little is known about how these models were built (e.g. the training data and its size are often not known), such as text-davinci-002. What we do know is that several of these models are deployed, either in external-facing commercial APIs (e.g. the OpenAI playground) or products (e.g. GitHub Copilot). That is, several of these models are having direct social impact at present. The absence of an evaluation standard compromises the community's ability to clearly and rigorously understand the overall landscape of language models. To demonstrate how uneven language model evaluation has been, we annotated the datasets used to evaluate more than 40 language models (i.e. all models evaluated in this work along with others like PaLM and Gopher) in Appendix F. We found major models such as T5 (11B) and Anthropic-LM v4-s3 (52B) were not evaluated on a single dataset in common in

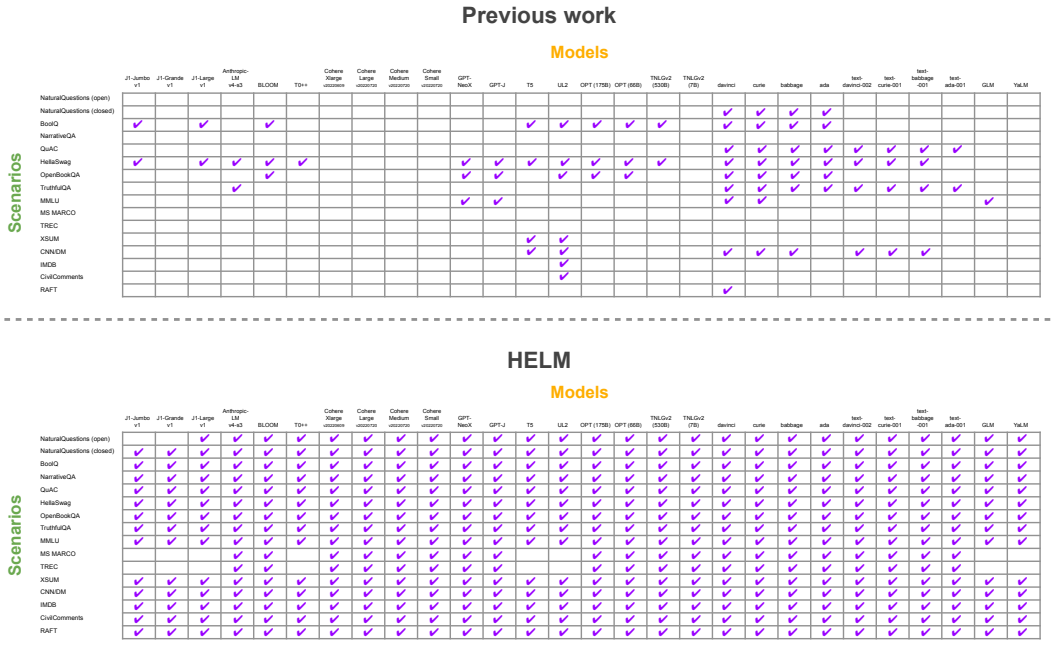

Figure 4: **Standardizing language model evaluation.** Prior to our effort (*top*), the evaluation of language models was uneven. Several of our 16 core scenarios had no models evaluated on them, and only a few scenarios (e.g. **BoolQ**, **HellaSwag**) had a considerable number of models evaluated on them. Note that this is *cumulative*: in the *top* plot, we not only document instances where the work introducing the model evaluated on a given scenario, but **any** subsequent work evaluated the model on the scenario (e.g. Tay et al. (2022a) in the paper on UL2 (20B) expanded the evaluation of T5 (11B) to include **HellaSwag** and several other datasets) under **any** conditions (e.g. fine-tuning, 0-shot prompting, 5-shot prompting). After our evaluation (*bottom*), models are now evaluated under the same conditions on many scenarios.

their original works (Raffel et al., 2019; Askell et al., 2021). In fact, several models (e.g. J1-Grande v1 (17B), Cohere xlarge v20220609 (52.4B), YaLM (100B)) do not report any public results prior to our effort (to our knowledge). And even for datasets that are frequently evaluated for across all 405 datasets evaluated in major language modeling works (e.g. **HellaSwag**; many of the datasets within GLUE and SuperGLUE), we find the evaluation conditions vary greatly. On **HellaSwag**, some prior work reports fine-tuned accuracies (e.g. T5 (11B)), whereas others report prompting accuracies (e.g. davinci (175B)).[4] Even when works report results through few-shot prompting, the exact details can vary, which in §8.2: PROMPTING-ANALYSIS we show leads to wild swings in accuracies (e.g. 30% to 80% for the same (model, scenario); see Zhao et al. (2021)).

In Figure 4, we make explicit how our evaluation changes the status quo. Previously, on average models were evaluated on 17.9% of our core scenarios, even after compiling evaluations dispersed across different prior works. We improve this to 96.0%.[5] By both evaluating these models on the same scenarios *and* by conducting the evaluation under standardized conditions (e.g. using the same few-shot prompting for all models), we facilitate direct head-to-head comparisons.

The importance of adaptation. To benchmark these models, we must specify an *adaptation* procedure that uses the general-purpose language model to tackle a given scenario (see Bommasani et al., 2021, §4.3). In this work, we adapt all language models through few-shot prompting, as pioneered by GPT-3 (Brown et al., 2020). Furthermore, we opted to choose relatively simple, generic prompts in order to orient the

[4] We emphasize our objective in raising this point is *not* to suggest that any individual work has evaluated improperly. In fact, for this case, few-shot prompting was not even popularized at the time of T5's writing. But, nonetheless, these models are evaluated under very different conditions (e.g. number of examples used in adaptation, ability to have white-box model access to use gradients to update the model), even if they are nominally evaluated on the same scenario.

[5] The remaining 4.0% is due to technical issues with specific models that we document in §6: MODELS.

development of language models towards generic language *interfaces* that respond robustly to direct natural language, rather than requiring model-specific incantations. Certainly stronger results could be obtained from more sophisticated prompting (e.g. chain-of-thoughts; Wei et al., 2022c), prompt decomposition (Wu et al., 2022; Press et al., 2022; Arora et al., 2022), and prompt-tuning (Lester et al., 2021; Li & Liang, 2021), potentially leading to qualitatively different findings (Suzgun et al., 2022). The exploration of adaptation strategies is another dimension of benchmarking which we leave to future work.

Caveats and considerations. Before presenting our empirical findings, we highlight three key considerations. First, while we standardize model evaluation, in particular by evaluating all models for the same scenarios, same metrics, and with the same prompts for 5-shot prompting, models themselves may be more suitable for particular scenarios, particular metrics, and particular prompts/adaptation methods. To be explicit, while some models may perform poorly under our evaluation, they may perform well in other contexts. Second, while the evaluation itself may be standardized, the computational resources required to train these models may be very different (e.g. resource-intensive models generally fare better in our evaluation), which is partially captured by our measurements of efficiency. Finally, models may also differ significantly in their exposure to the particular data distribution or evaluation instances we use, with the potential for *train-test contamination*. We emphasize that we have a limited understanding on how contaminated models are, and to what extent this compromises the validity and legitimacy of our evaluation, though we do provide all evidence we are aware of in Appendix G.

1.2 Empirical findings

To give a sense of the magnitude of our evaluation, we ran a total of 4,939 runs (i.e. evaluating a specific model on a specific scenario). This amounts to a total cost of 12,169,227,491 tokens and 17,431,479 queries across all models, $38,001 for the commercial APIs, and about 19,500 GPU hours worth of compute for the open models.

Here is a summary of the high-level findings:

1. **The benefits of instruction-tuning.** Across the core scenarios, we find that text-davinci-002 performs best on our accuracy, robustness, and fairness metrics, with Anthropic-LM v4-s3 (52B) being in the top 3 for all 3 metrics (despite being more than 10× smaller in model scale compared to TNLG v2 (530B), which is the second most accurate and fair) as shown in Figure 26. Given the very strong performance of both models, and that they are the only instruction-tuned models we evaluate (beyond the much smaller OpenAI model variants), this suggests instruction-tuning provides a broad set of advantages.

2. **Relating model accuracy with model access.** In light of the high accuracies of Anthropic-LM v4-s3 (52B) (closed), TNLG v2 (530B) (closed), and text-davinci-002 (limited-access), we observe a consistent gap on all core scenarios (Figure 28) between the current open models and non-open models. We emphasize that this gap reflects the current snapshot of models we evaluate (Table 5), and that the gap could grow or shrink over time as new models are released. On one hand, we see the recent release of open models (OPT (175B), BLOOM (176B), GLM (130B)) as greatly reducing the gap over the past year, but we also have not evaluated some non-open models (e.g. PaLM, Gopher) that we expect to be quite accurate. In either case, monitoring this gap over time is crucial for tracking the accessibility (or lack thereof) and ultimately the power dynamics associated with language models.

3. **Calibration.** We observe that the relationship between accuracy and calibration (§4.4: METRICS-CALIBRATION) depends on the scenario and adaptation procedure (Figure 24, Figure 25). As an example, for **HellaSwag**,[6] improving accuracy worsens calibration, whereas for **OpenBookQA**,[7] improving accuracy improves calibration.

[6]See `https://crfm.stanford.edu/helm/v0.1.0/?group=hellaswag`.
[7]See `https://crfm.stanford.edu/helm/v0.1.0/?group=openbookqa`.